Olfactory & Taste Sensation Notes

Disturbance in Taste Sensation

- Dysgeusia (Dysgueria): Disturbance in taste sensation [cite: 1].
- Hypogeusia (Hypogueria): Decrease in taste sensation [cite: 1].

Sense of Smell (Olfaction)

- Originates in the nasal cavity [cite: 1].
- Olfactory receptors are present on the olfactory mucous membrane that lines the upper part of nostrils [cite: 1].
- Olfactory mucous membrane contains 10 to 20 million olfactory receptor cells, supported by sustentacular cells [cite: 1].
- Olfactory mucous membrane also contains mucus-secreting Bowman's glands [cite: 1].
- Olfactory receptor cells contain bipolar neurons [cite: 1].
- Dendrite of this neuron is short and has an expanded end called an olfactory rod [cite: 1].
- From the olfactory rod, 10-12 non-myelinated cilia arise [cite: 1].
- Cilia project into the nasal mucus membrane [cite: 1].
- Mucus secreted from the Bowman's gland continuously lines the olfactory mucus membrane [cite: 1].
- This mucus contains specialized proteins which assist the action of odoriferous substances on olfactory receptors [cite: 1, 2].
- Olfactory nerve is the 1st cranial nerve [cite: 2].
- Sensory receptors and fibres originate in the upper part of the nasal mucus membrane [cite: 2].
- Passes through the ethmoid bone to the olfactory bulb, then finally to the temporal lobe of the cerebral cortex where the area of perception of smell takes place [cite: 2].

Olfactory Transduction

- Process of conversion of chemical energy by olfactory receptor to action potential [cite: 2].
- Odoriferous substance dissolves in the mucus covering the nasal mucus membrane [cite: 2].
- Molecule of dissolved substance binds to receptor proteins in cilia [cite: 2].
- Substance-receptor complex forms [cite: 2].
- Activation of adenylyl cyclase enzyme [cite: 2].
- Formation of c.AMP [cite: 2].
- Activation of Na^+ channels [cite: 2, 3].
- Influx of Na^+ ions [cite: 3].
- Development of receptor potential in olfactory receptor [cite: 3].
- Development of action potential in olfactory sensory nerve [cite: 3].

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Disorders of Nose

- **1. Anosmia:** Total loss of sensation of smell; inability to detect/recognize any odour. May be temporary or permanent. Caused by blockage during cold/allergic reactions, tumours/cysts in olfactory tract, meningitis, Parkinson's, or Alzheimer's diseases [cite: 3].
- **2. Hyposmia:** Reduced ability to recognise/detect any odour. The odour can be detected only at higher concentration [cite: 3].
- **3. Hyperosmia:** Exaggerated sensation of smell (commonly seen during brain injury, epilepsy) [cite: 3].

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